

Ad Hoc Networks

Advanced Mobile Communication Networks



Basics

Questions to be answered:

- What are the various forms of ad-hoc networks?
- How are ad-hoc networks formed? What are their constituent components?
- What characteristics are exhibited by ad-hoc networks?
- What advantages do ad-hoc networks offer?
- Which applications can benefit from the use of ad hoc networks?
- Discuss the advantages and disadvantages of different routing strategies for ad hoc networks
- What is the idea of delay-tolerant networking?
- What are the advantages and disadvantages of geographical routing

Ad Hoc Networking Applications

Military



Automotive networks



Interactive lectures

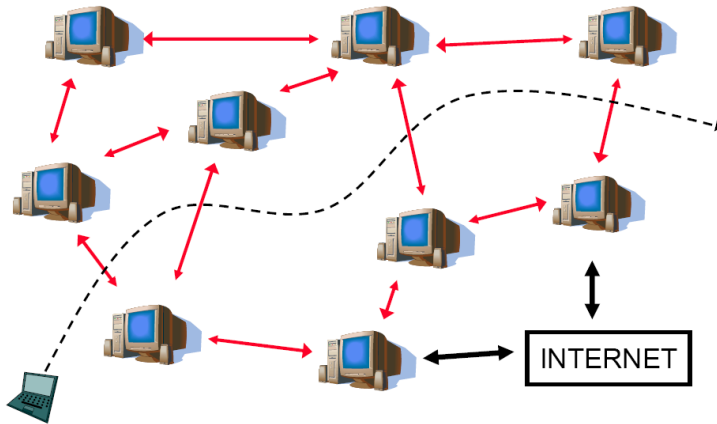


Example applications:

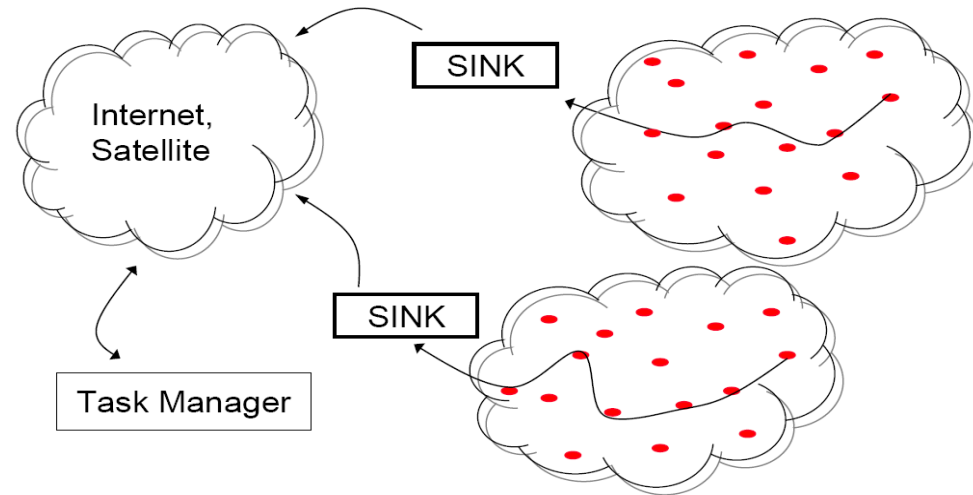
- Military, e.g. battlefield communication
- Human gatherings, like conferences, exhibitions, meetings, lectures, gaming
- Car-to-car and car-to-X communication, e.g. accident warnings, traffic control
- Sensor applications, e.g. for fire detection, earthquake warnings, etc.
- Extension of cellular networks beyond infrastructure coverage
- Disaster communications, e.g. after earthquakes, tsunamis, terrorist attacks, ...
- Delay-tolerant networking, e.g. wildlife tracking, long-term data collection

Ad Hoc Networks Types

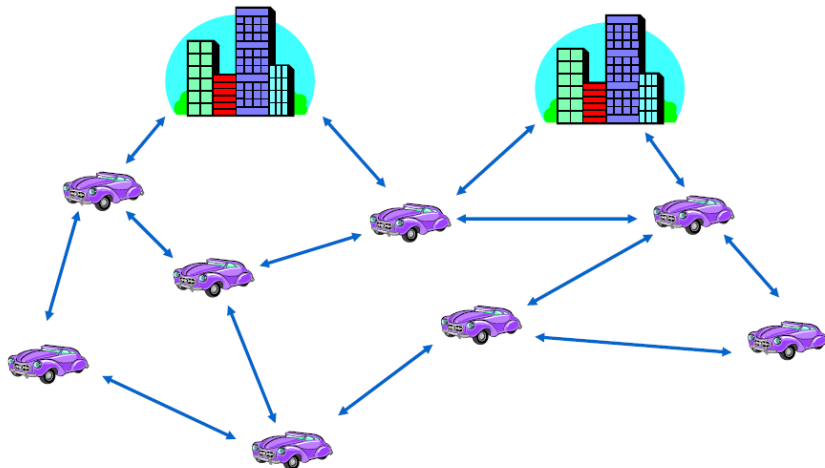
Wireless Mesh Networks (WMNs)



Wireless Sensor Networks (WSNs)



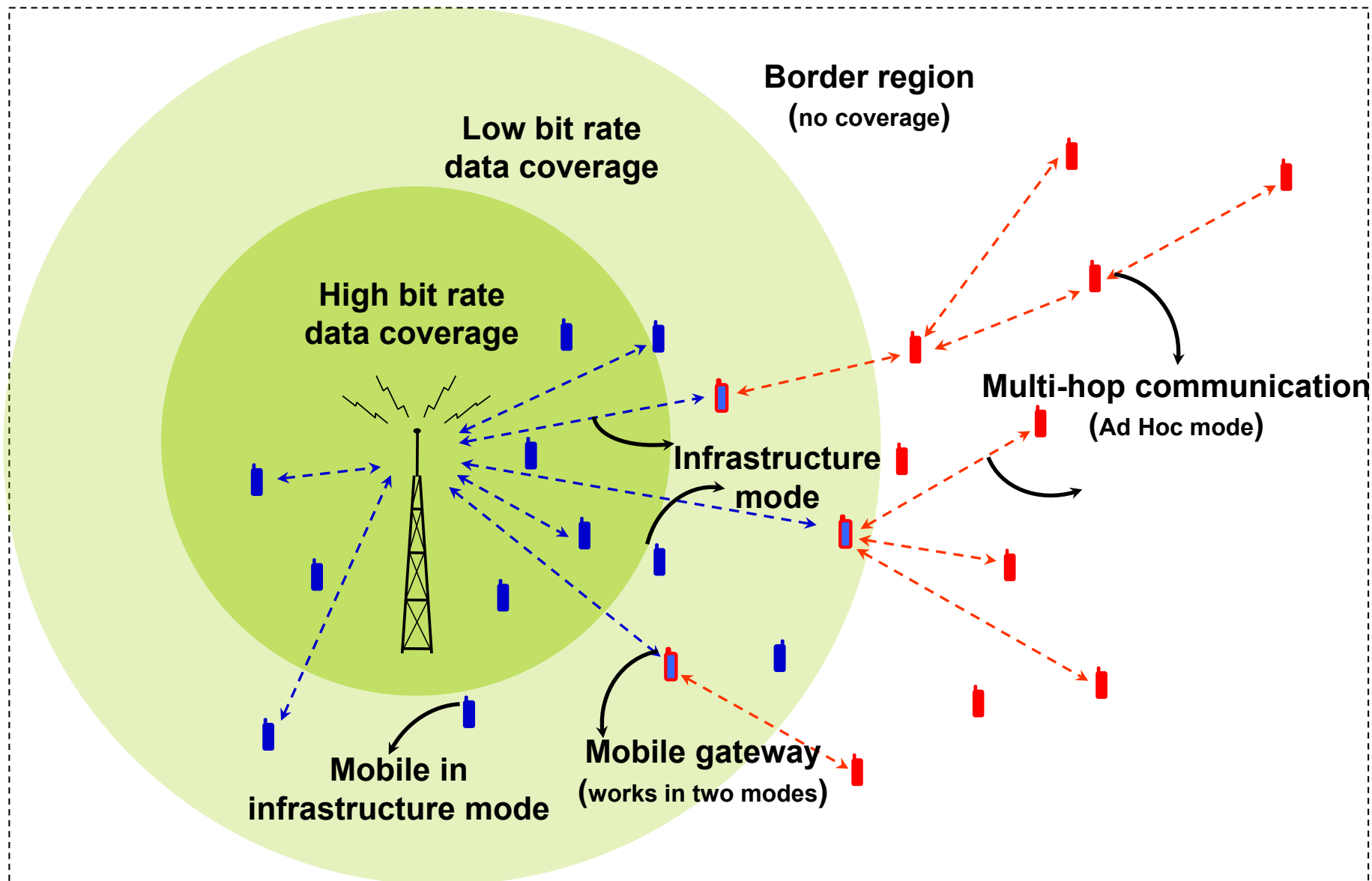
Vehicular Ad Hoc NETWORKS (VANETs)



Delay-Tolerant Networks (DTNs)

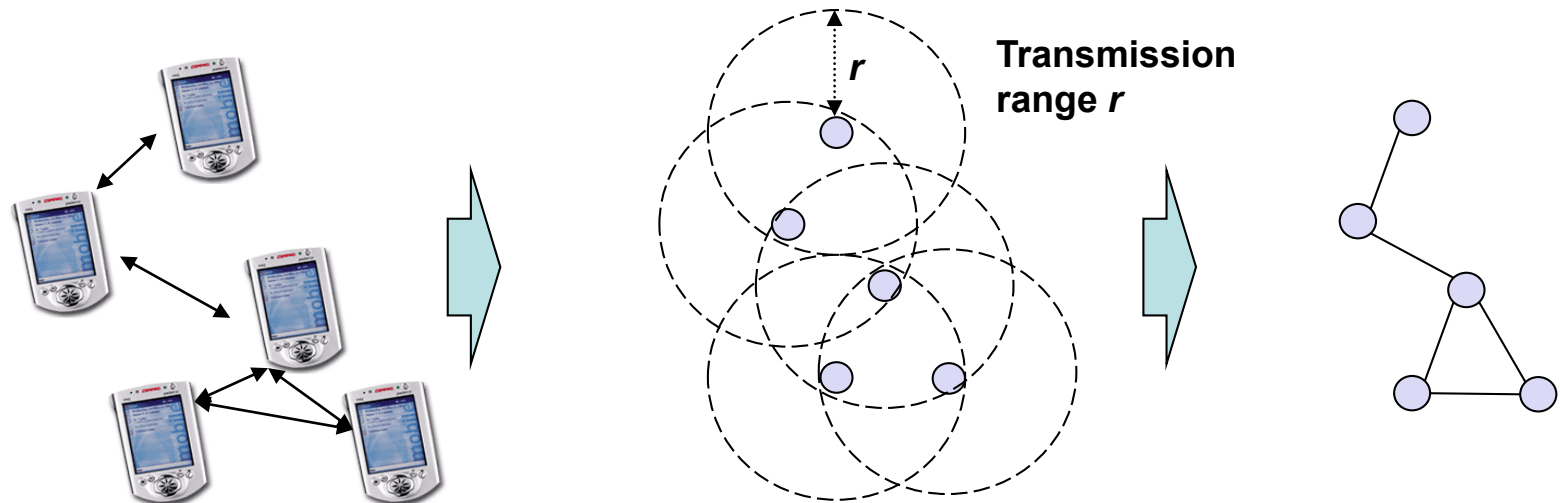


Ad Hoc Networks Types: Extension of 4G/5G Networks



Mobile Ad Hoc Networks (MANETs)

- Spontaneous federation of wireless devices
 - No infrastructure (base station / access point), no backbone
 - Devices can be mobile
- Packet-based forwarding
 - Mobile device also serve as routers
 - Routes between devices can span multiple hops
- Ad hoc networks are self organizing
 - No central components



MANET Challenges

- Highly dynamic topology
 - Mobility of devices
 - Varying quality of wireless channel (fading)
 - Partitioning and merging of ad hoc networks possible
- Limited battery capacity of mobile devices
 - Additional battery drain due to (e.g.) routing functionality
- Limited bandwidth
 - Additional bandwidth required for routing and MAC functionality
- Wireless medium is broadcast medium
 - Hidden and exposed terminals
- Asymmetric/unidirectional links
 - Different quality in both directions
- Time synchronization difficult
 - Problem for low power modes (e.g. sleeping periodically)
- Security mechanism hard to apply
 - Every devices must be able to forward packets → no encryption of routing headers

⇒ **Challenge of route maintenance**

MANET Routing

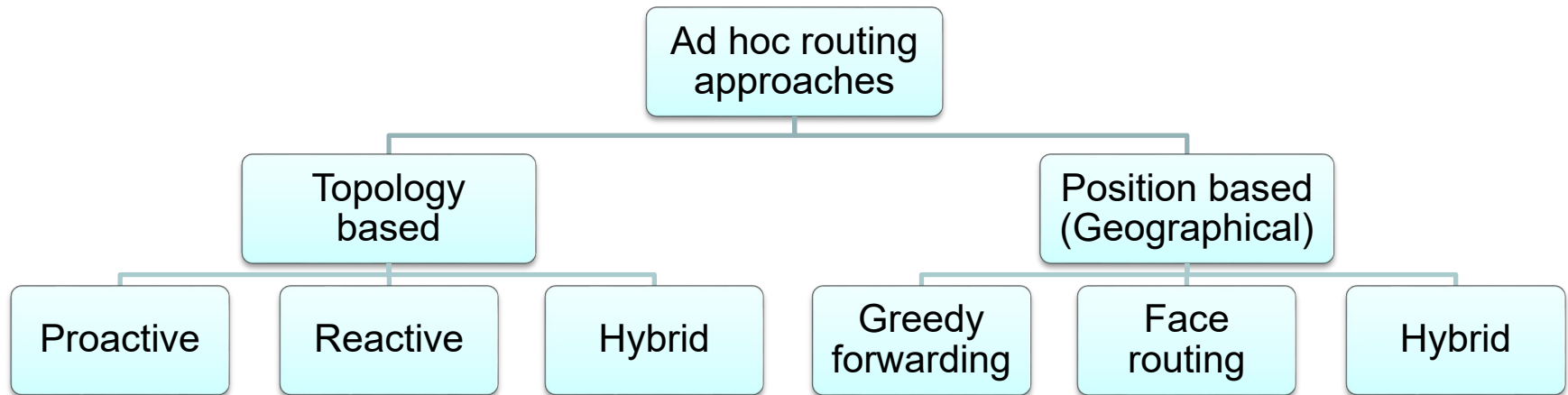
Questions to be answered:

- What are the challenges of routing in multi-hop ad-hoc networks?
- What are the differences between link state and distance vector routing protocols?
- What are different classes of ad hoc routing protocols?
- What are differences between reactive and proactive routing protocols?
- What is the idea of greedy geographical routing?
- What types of information are needed to apply geographical routing protocols in ad hoc networks?

MANET Routing

- Protocols for wired networks (e.g., RIP, OSPF) have to be adapted
 - to provide faster convergence and adapt to changes
 - to reduce overhead
- Possible metrics for “shortest path”
 - Minimum number of hops
 - Minimum delay
 - Minimum packet loss probability
 - Minimum congestion (load balancing)
 - Minimum interference
 - Maximum signal stability, stable route
 - Maximum battery lifetime of mobile device
 - Maximum lifetime of entire network
 - e.g. until network is partitioned due to nodes running out of power

Classification of MANET Routing Protocols

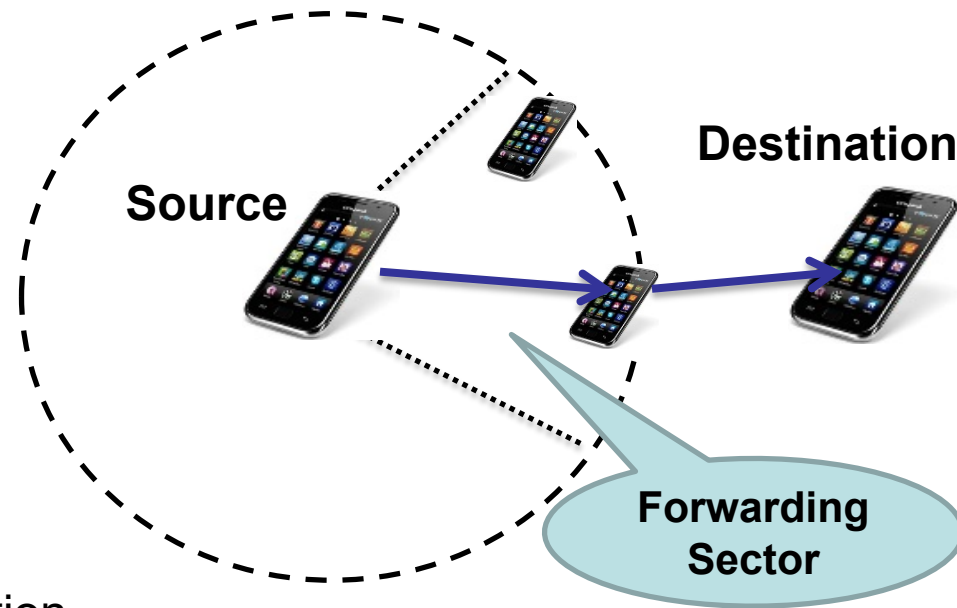


Topology-based routing:

- Proactive or table-driven (Examples: DSDV, GSR, WRP, OLSR)
 - Routes are calculated periodically and before needed
 - Keep routing information in all nodes up-to-date
- Reactive or on-demand (Examples: AODV, DSR, LMR, ABR)
 - Routes are only calculated when needed
 - Do not keep routing information in all nodes up-to-date
- Hybrid (Examples: ZRP, SHARP, Safari)

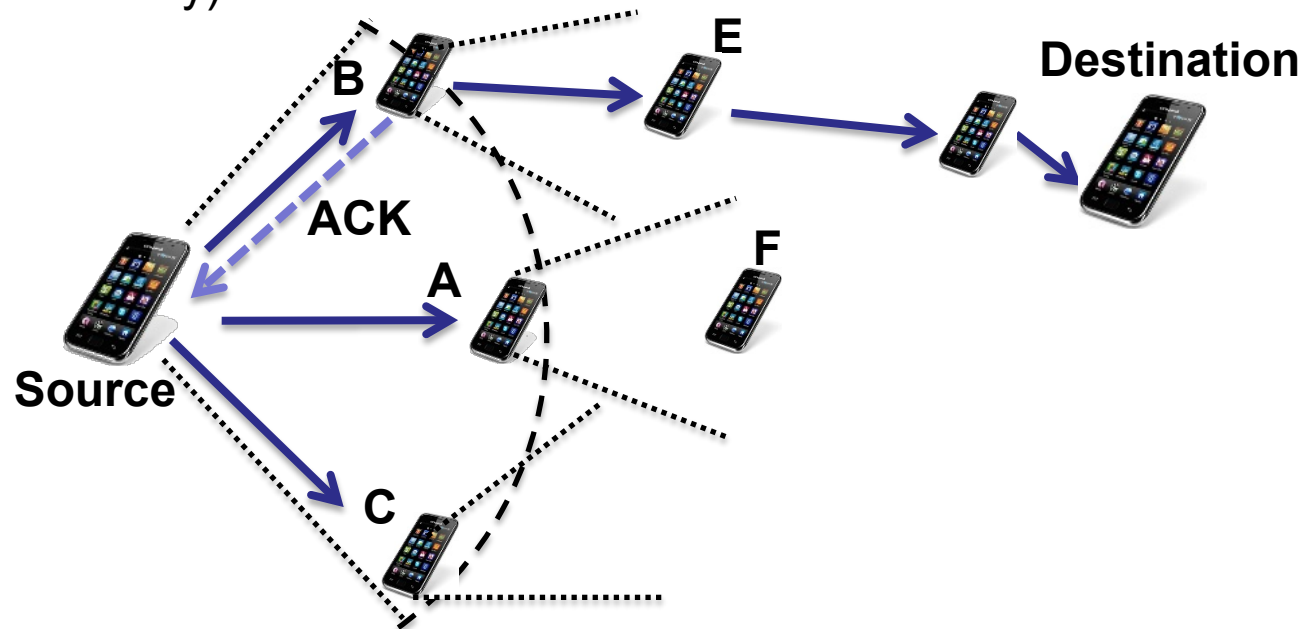
Geographical Routing

- Based on location information
 - GPS information
- Forwarding sector
 - Candidate neighbors
- Hop-by-hop decision making
 - No need for topology information
 - Based on location information of destination and intermediate nodes
 - Assumption: destination location must be known to the source
- Local information
 - Obtained periodically or on-demand



Opportunistic Geographical Routing

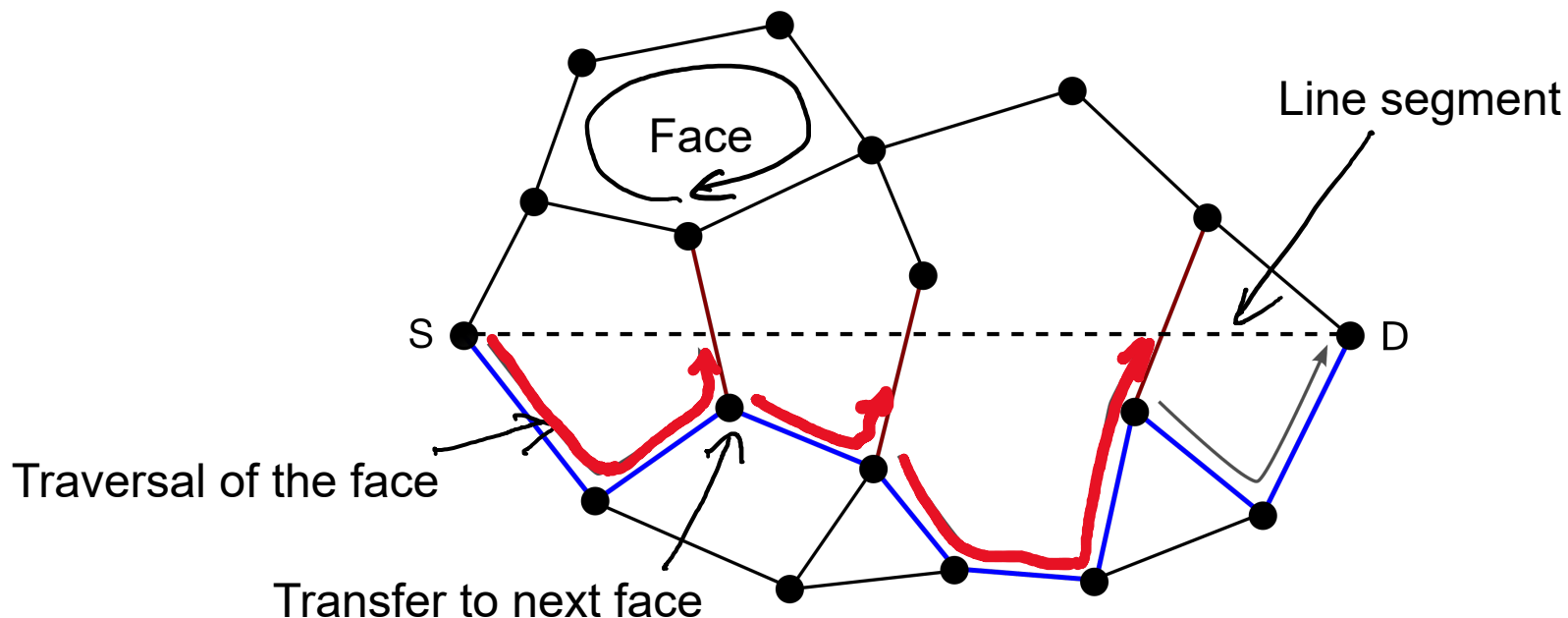
- Prioritization of candidates in source (or intermediate nodes)
 - Utility function
 - QoS requirements
 - Beacon information (distance and load)
- Decision in candidates
 - Environmental conditions (error rate and mobility)



Geographical Face Routing

Algorithm (simplified, lots of variants exist):

- (1) Forwarding is along the boundary of the “face” (a circle defined by a set of nodes) until the intersection with the line segment (dotted line) is hit
- (2) Instead of surpassing the line segment, the next face is traversed till the line segment is hit again
- (3) Repeated until destination is reached



Delay-Tolerant Networking

Idea: Nodes are not continuously connected to each other, rather use communication opportunities due to the mobility of nodes (message ferries)

- Message ferries
 - Message ferries may be cars, trains, UAVs, animals, smartphones
 - Message ferries mobility may be known (bus network with fixed schedule), partially known (animals or people following some routines) or just stochastically known
 - Mobility of nodes may also be controlled by the need for message transfers, e.g. special UAVs
- Message traffic
 - Message traffic may be totally random or follows some pattern
 - Messages are assumed to be tolerant to delay
- Forwarding between message ferries (and possibly other nodes) may be done as unicast or multicast/broadcast depending on requirements on reliability, average delay, etc.

Proactive Routing Protocols

Idea: nodes constantly construct and maintain routes to all other nodes

Review from routing in wired networks:

- **Distance Vector Routing**

- Each node computes for each destination in the network (stepwise starting with its neighbors) a matrix with an entry for each pair of destination and next hop with the total length of the path to the destination via this next hop
- Matrix (the complete routing table) is updated and sent to all neighbors whenever an update from neighbors arrives
- In conclusion, each node shares its complete (huge) routing table with its neighbors
- Examples
 - Wired networks: Routing Information Protocol (RIP)
 - Ad hoc networks: Destination-Sequenced Distance Vector Protocol (DSDV)

- **Link State Routing**

- Each node distributes its link state (its neighbors) to all nodes in the network
- Receiving this (small) information from all nodes each node knows the entire network topology
- Examples
 - Wired networks: Open Shortest Path First (OSPF)
 - Ad hoc networks: Optimized Link State Routing (OLSR)

Optimized Link State Routing (OLSR)

Idea: Link State routing

- Regular exchange of neighbor information (1 and 2 hops) with all other network nodes
- From this information each node maintains a topology graph from which routing table with all known destinations in the network is computed

Tasks:

- Link sensing & neighbor sensing
 - Nodes periodically exchange “Hello” messages to detect neighbors
 - “Hello” messages permit nodes to learn about neighborhood up to 2 hops
- Multi-Point Relay (MPR) selection
 - Based on “Hello” messages, each node selects a set of its neighbors as MPRs
- Topology information diffusion
 - MPRs receive and forward Topology Control (TC) messages whenever there is a topology update
 - MPRs reduce overhead by reducing the number of nodes flooding the network

Reactive Routing – AODV

Questions to be answered:

- Which routing table entries are established with the RREQ and RREP messages, respectively?
- How is the forwarding and how the reverse path established in the routing tables?
- What is the purpose of the sequence number and the hop count maintained in the routing tables?
- What is the purpose sequence number in RREQs and RREPs?
- How are multiple RREQs handled by the receiving node? What is the meaning of sequence number and hop count for this handling?
- How are link failures detected? What are the mechanisms to repair the link or route?
- What is the purpose of the RERR message? Which routing tables are modified with this message?

Reactive Routing Protocols

Idea:

- Nodes routing tables are built on demand, and only contain forwarding information for currently used destinations
- No periodic route maintenance

Tasks:

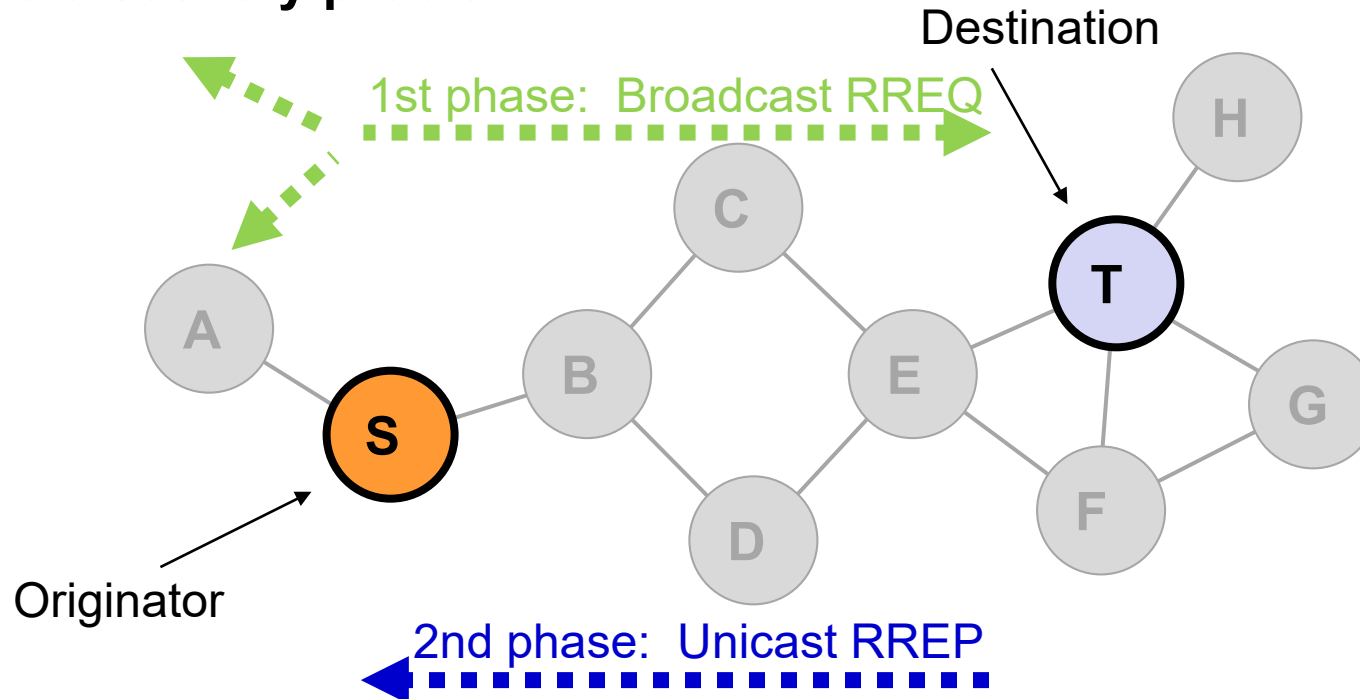
- Route discovery triggered whenever route to destination is unknown
- Route maintenance employed for routes that are currently in use

Comparison to proactive protocols

- Advantages
 - No unnecessary construction & maintenance of routes
 - No periodic messages → lower resource consumption
- Disadvantages
 - Delay at the beginning of communication due to route discovery
 - Control overhead depends on number of connections and mobility

Ad-hoc On-Demand Distance Vector Routing (AODV)

Route discovery problem



Two phases:

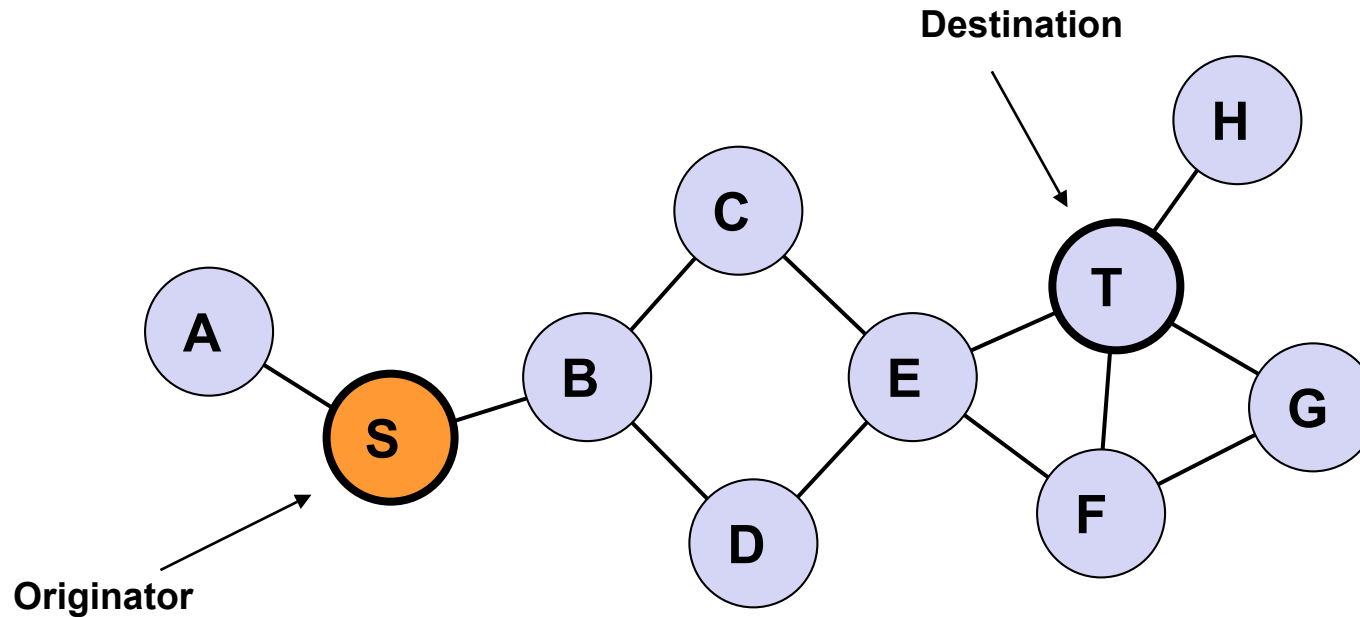
- Establishing reverse path T to S with Route Requests (RREQs)
- Establishing forwarding path from S to T with unicast Route Reply (RREP)

AODV Route Discovery – Reverse Path Establishment

Route discovery: Route REQuest (RREQ)

- Sender S floods RREQ (broadcast) for destination T
 - RREQ contains
 - source and destination addresses
 - RREQ ID (incremented by the sender for repeated RREQ after timeout to discriminate from earlier attempts)
 - source address + RREQ ID => unique ID
 - source and destination sequence numbers (to ensure "freshness" of routing information and to avoid forwarding loops)
 - hop count (incremented by each intermediate node before forwarding RREQ)
- Whenever a node receives a RREQ from a neighbor it saves this routing information and generates a respective routing table entry for the **construction of the reverse path** from destination T to source S (Only works for bidirectional links!)

AODV: Route Discovery – Flooding of RREQs

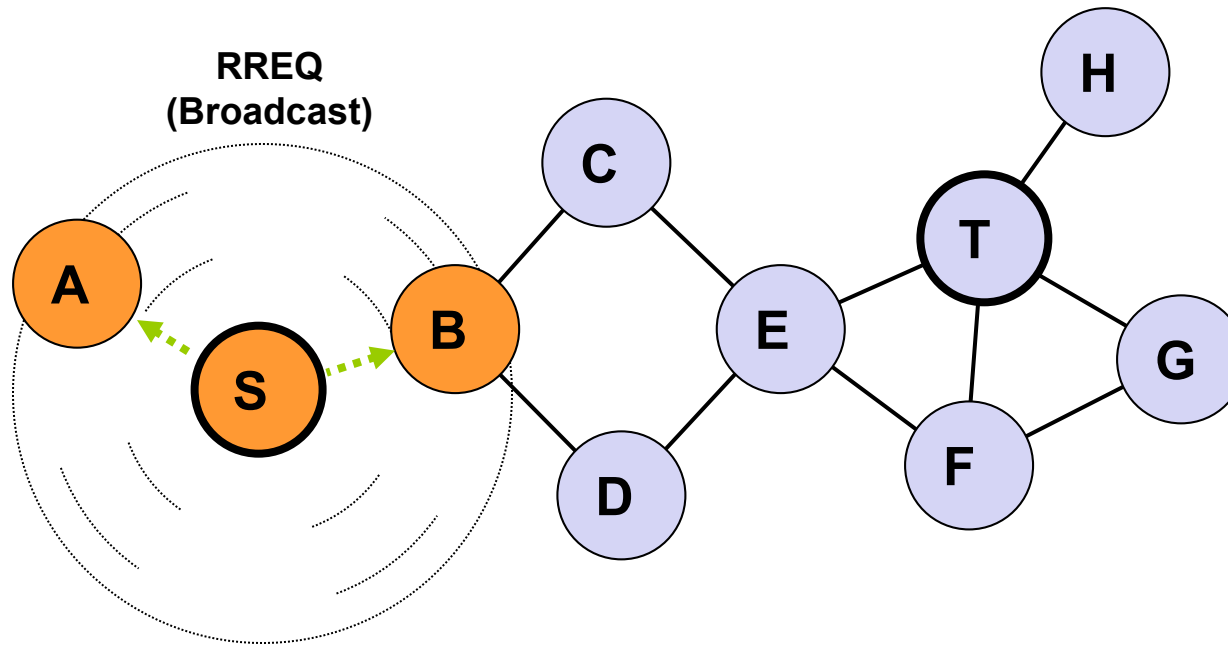


Nodes that have already received RREQ



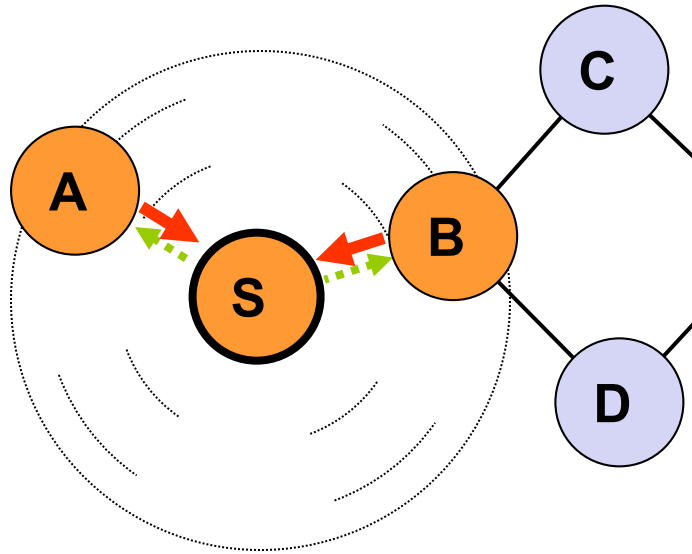
Nodes in mutual transmission range (bidirectional links)

AODV: Flooding of RREQs



- RREQ is transmitted by broadcast
 - Received by all nodes in transmission range

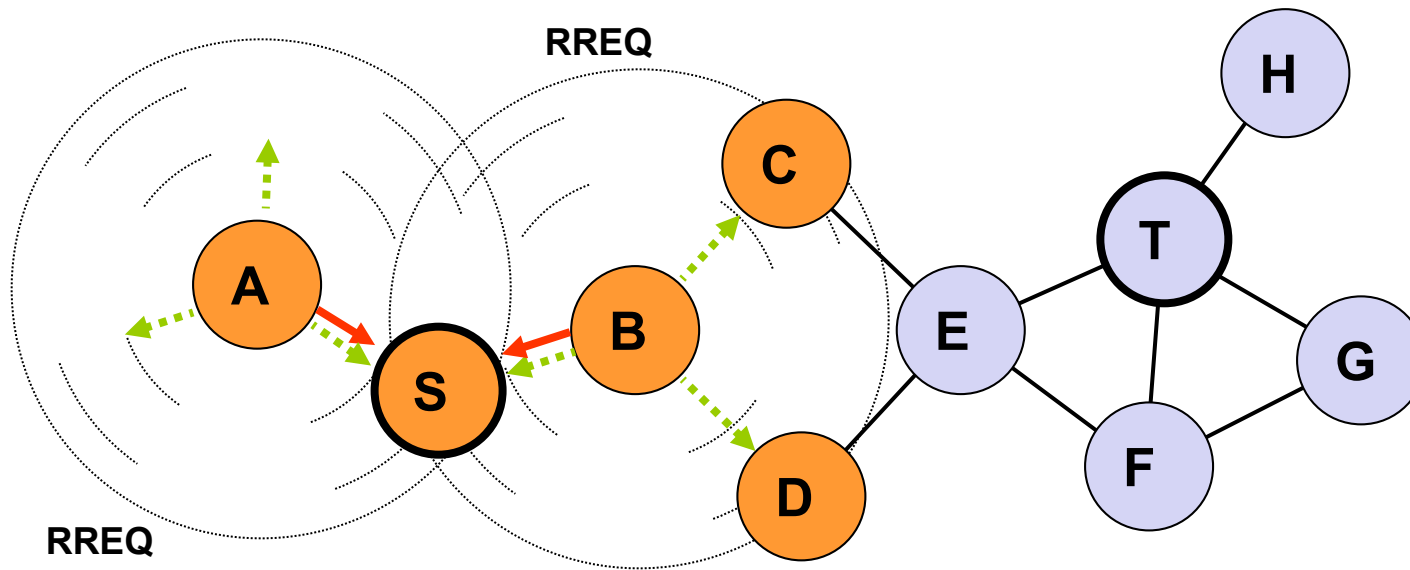
AODV: Flooding of RREQs



- Upon receipt of RREQ, receiving node updates its routing table (→)
- Table entries:
 - Destination address
 - Dest. sequence number
 - Next hop address
 - Hop count
 - Time to live

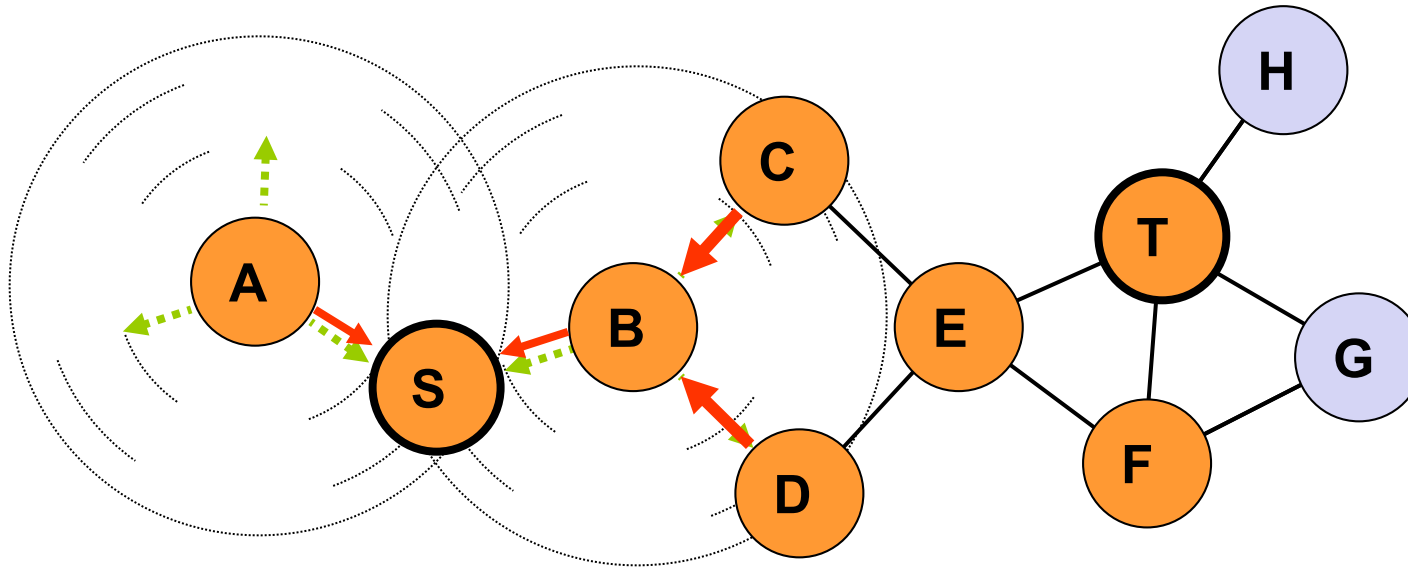
→ Routing table entries in intermediate node for source of RREQ S

AODV: Flooding of RREQs

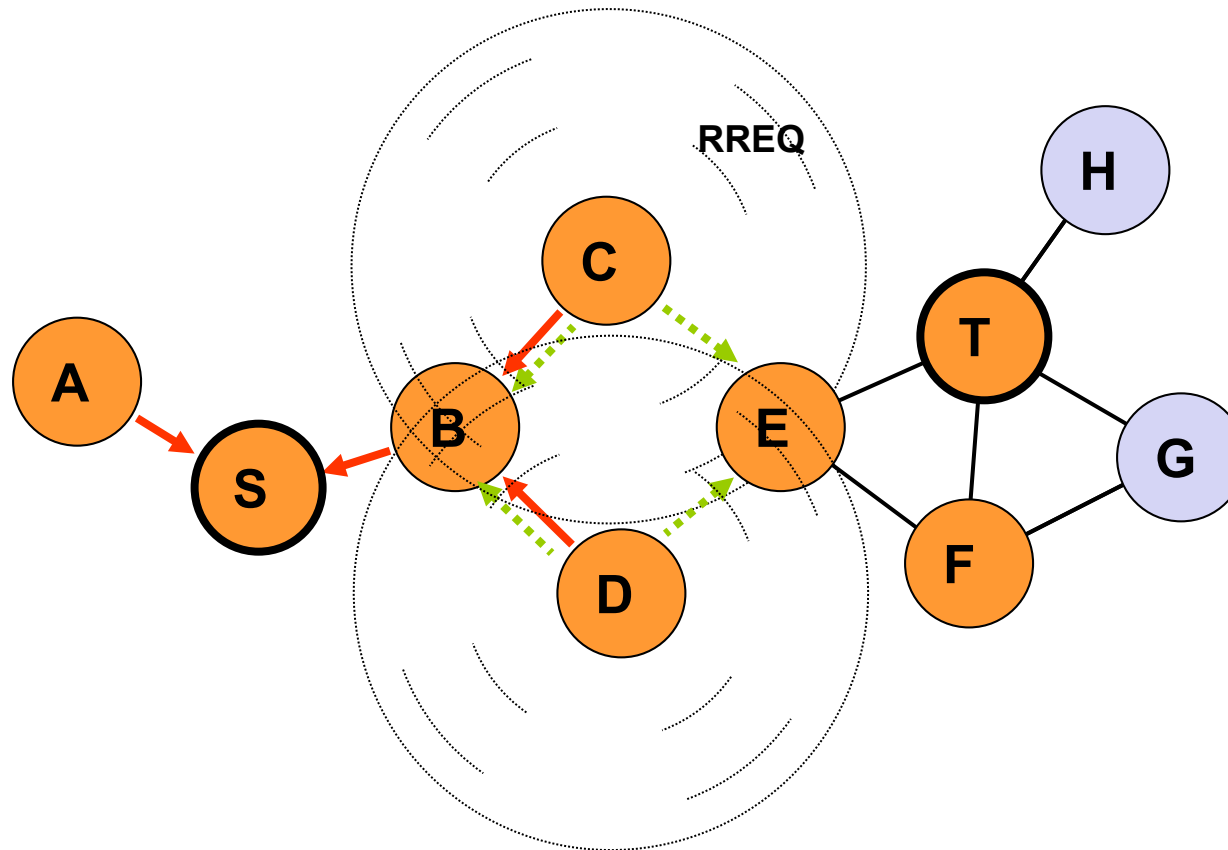


AODV: Flooding of RREQs

→ Constructed reverse path

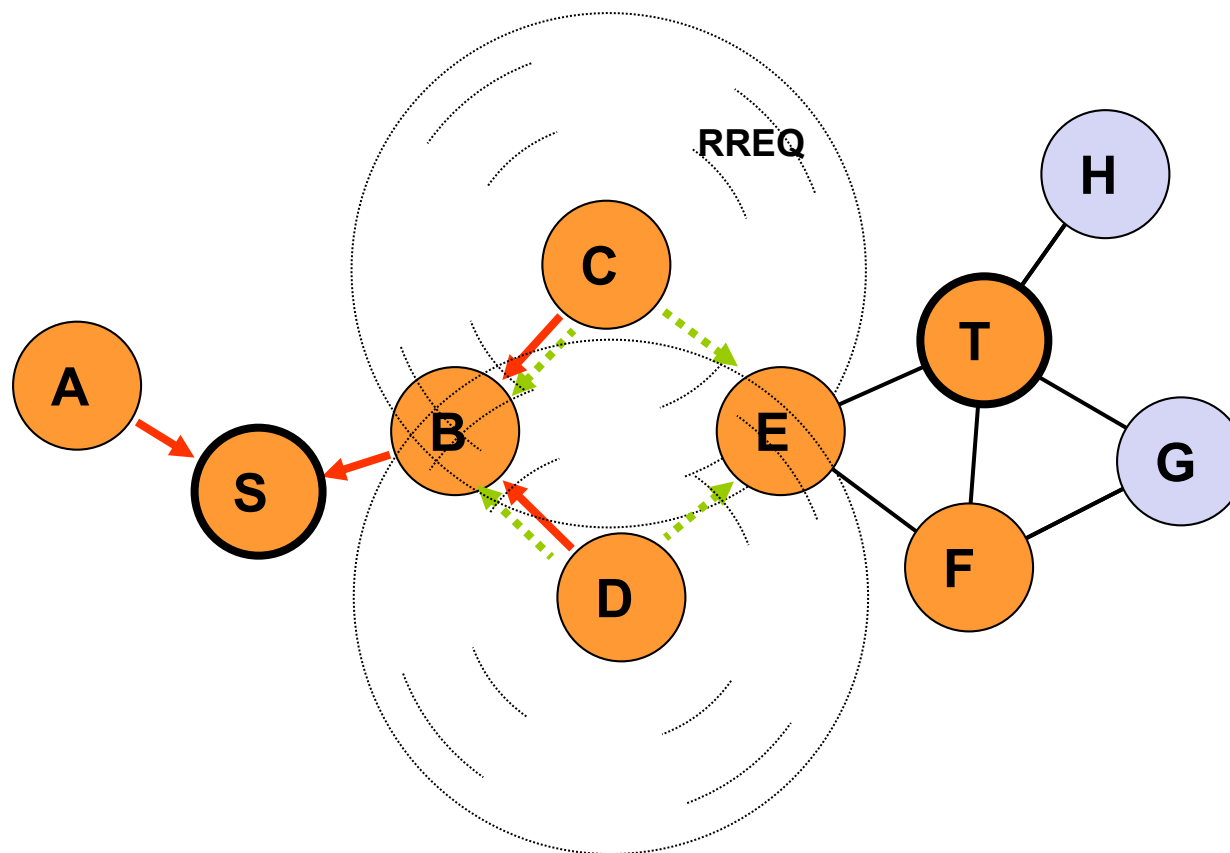


AODV: Flooding of RREQs



- Cycling RREQs (in B) are detected by hop count and discarded
- Outdated RREQs are detected by old sequence numbers and discarded

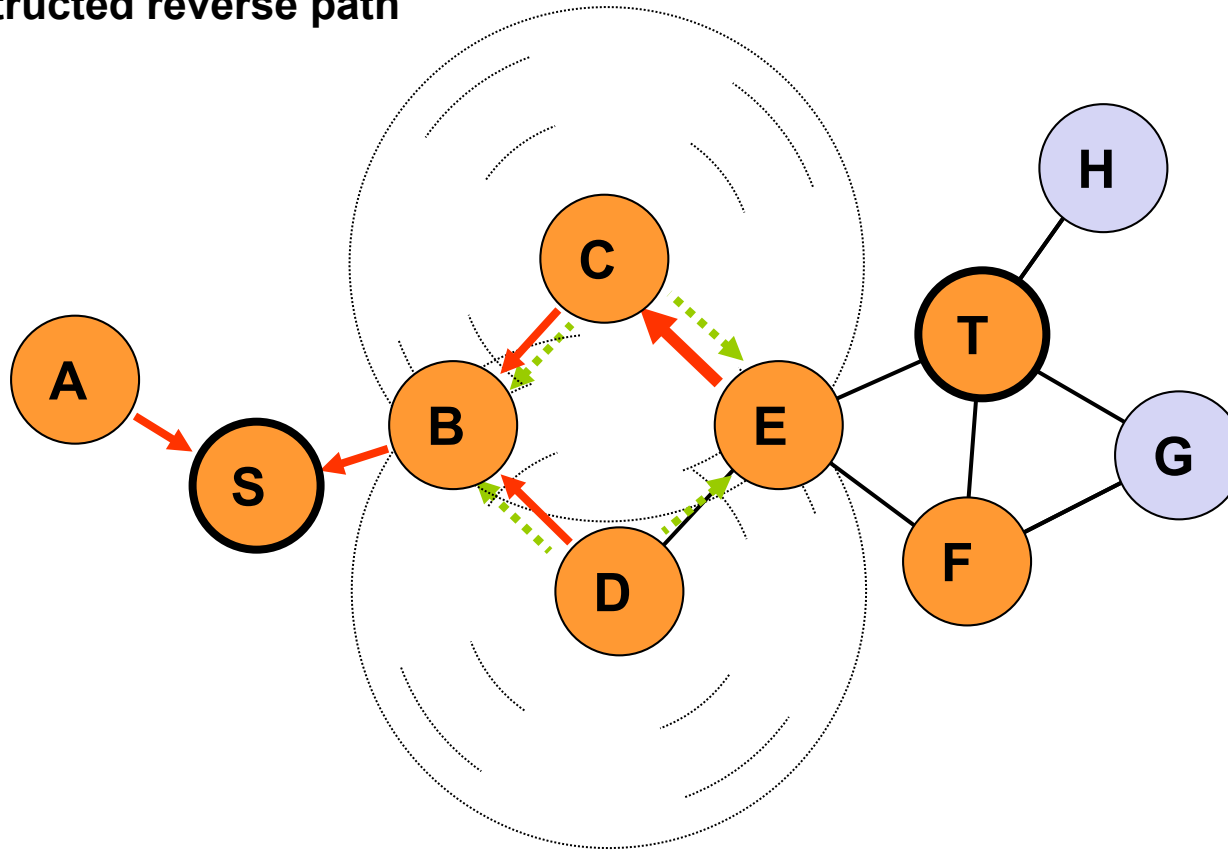
AODV: Flooding of RREQs



- Duplicate RREQ (in B and E) are detected by sequence numbers and discarded

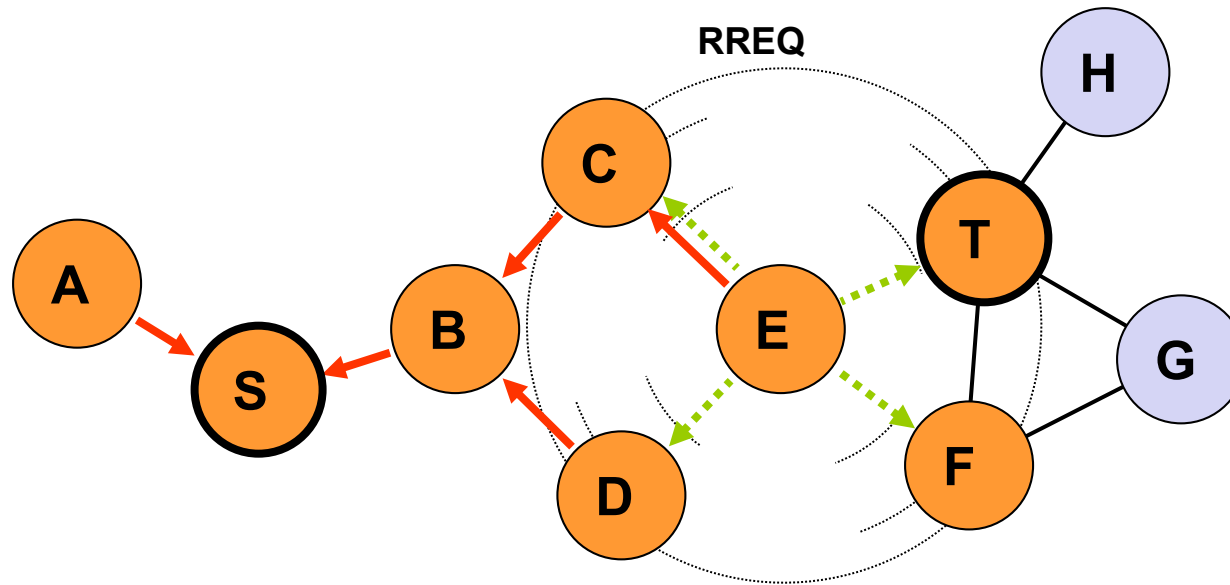
AODV: Flooding of RREQs

→ Constructed reverse path



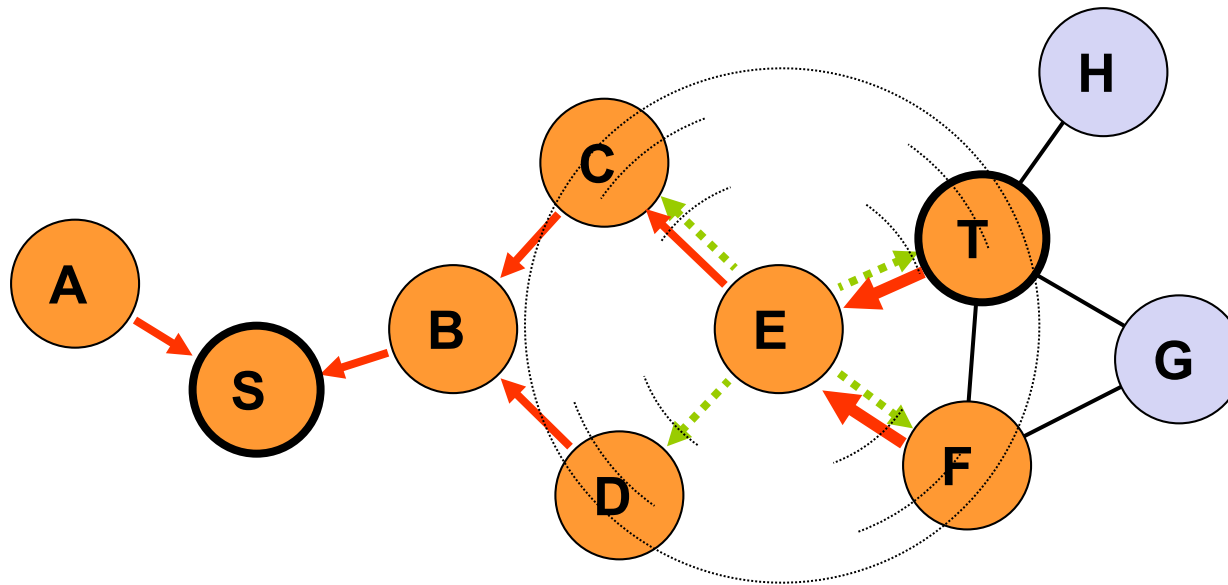
AODV: Flooding of RREQs

→ Constructed reverse path



AODV: Flooding of RREQs

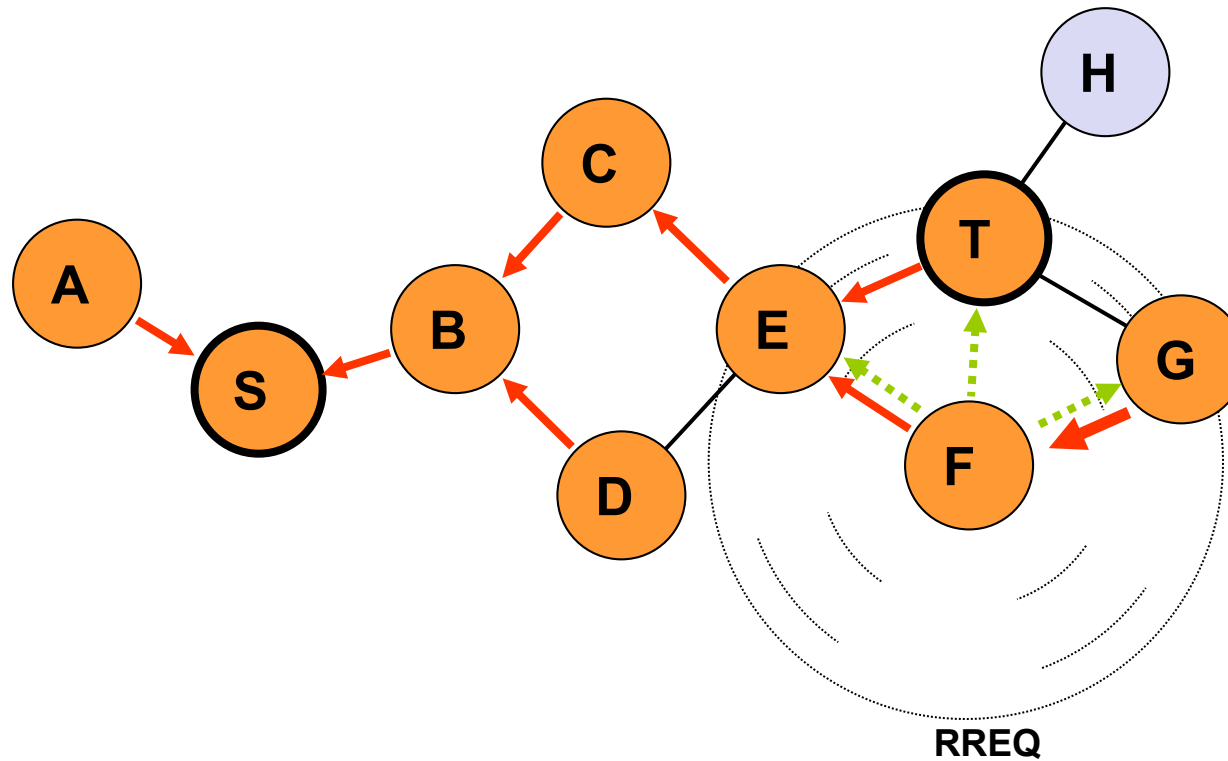
→ Constructed reverse path



- Destination T has received the RREQ
 - Complete reverse path from T to S has been constructed

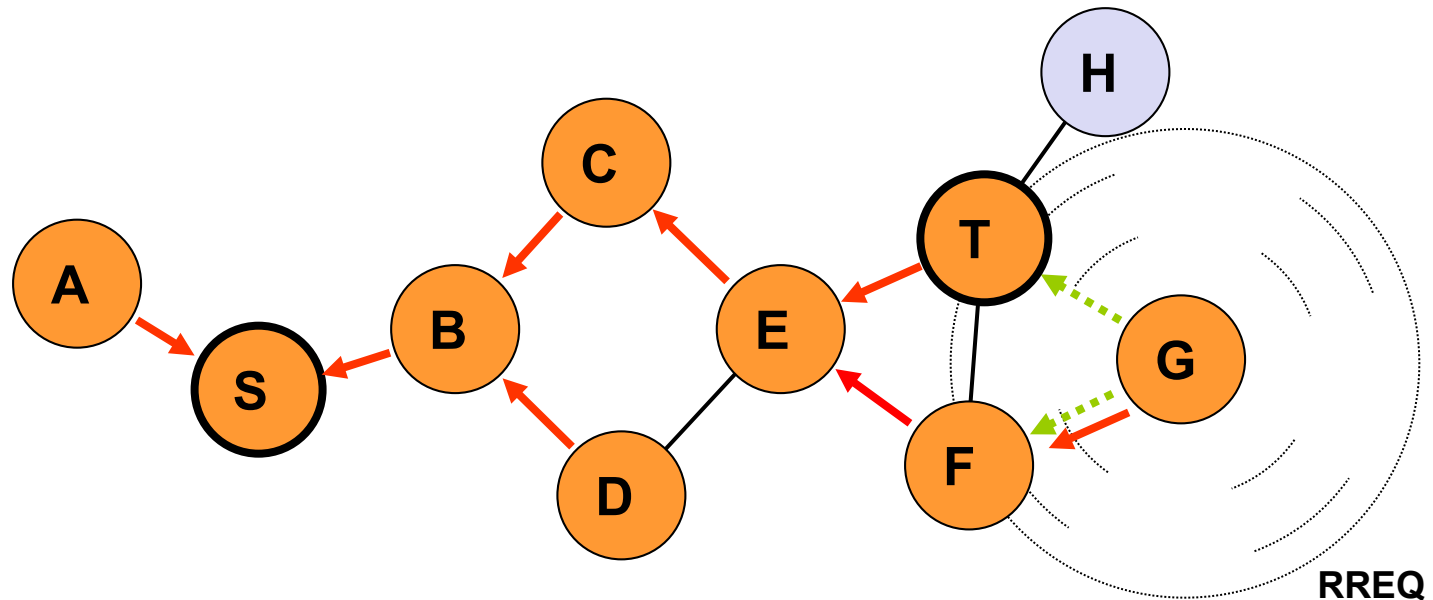
AODV: Flooding of RREQs

→ Constructed reverse path



AODV: Flooding of RREQs

→ Constructed reverse path



- Node T replies with RREP to S using reverse path constituted by the individual routing table entries established with RREQs
 - Using these table entries each hop on reverse path can be directly addressed (no broadcast)

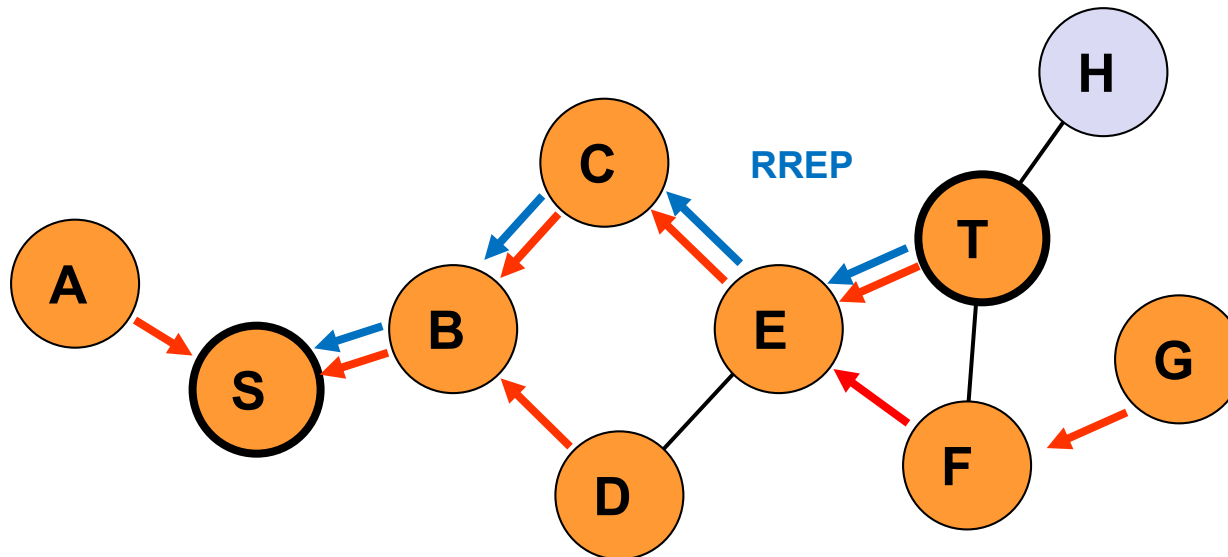
AODV Route Discovery – Forward Path Establishment

Route discovery: Route REPlY (RREP)

- When RREQ reaches destination T (or an intermediate node that has valid routing information towards T) a RREP is generated
 - RREP contains
 - RREQ originator address and destination addresses
 - destination sequence number
 - Lifetime of route
- RREP is forwarded towards S on reverse path established by RREQs
- The information in the RREP is used for the generation of a routing table entry for the destination T in every traversed node, thus **constructing the forward path** from source S to destination T
- Forward and reverse path can now be used for the exchange of data packets between S and T

AODV: Route Discovery – Unicast RREPs

→ Constructed reverse path

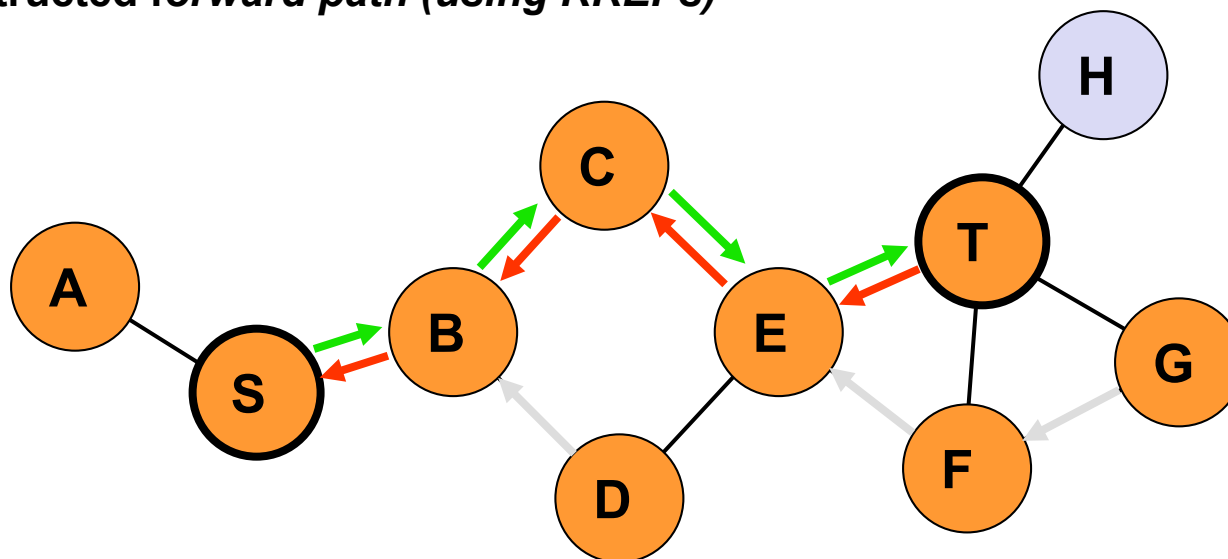


- Node T replies with RREP to S using reverse path constituted by the individual routing table entries established with RREQs
 - Using these table entries each hop on reverse path can be directly addressed (no broadcast)

AODV: Unicast RREPs

→ Constructed reverse path (using RREQs)

→ Constructed forward path (using RREPs)



- RREP updates the routing table of each node, thus establishing the forward path (→) from S to T
- Unused reverse path (→) entries will be discarded after timeout
- Data exchange from S to T starts over the constructed path after successful route discovery

AODV: Sequence Number Details

Idea:

- Each node maintains an individual sequence numbers (SQN) denoting the freshness of its routing information in the networks routing tables

Routing table entries:

- In the routing tables, each destination address entry is associated with a destination SQN denoting its freshness

Dest. addr.	Dest. SQN	Next Hop	Hop Count	Life time
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SQN updates:

- A node increases its own SQN whenever a new route needs to be established (route discovery) or updated (route maintenance), e.g. due to link changes or outdated routing information

SQN updates in routing tables:

- Routing entries related to a destination address are updated whenever a RREQ or RREP contains a newer SQN for this destination (independent of hop count!)

AODV: Sequence Number Details

SQN update details:

- An originating node increases its own SQN whenever it initiates a new route discovery (RREQ)
- A destination receiving a RREQ also increases its SQN before replying with a RREP



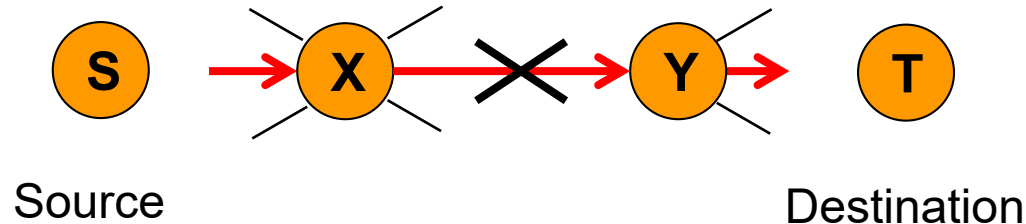
Use of SQN by signaling messages:

- The SQN of originator (S) and destination (T) are included in each RREQ and RREP
- The originator SQN is used to update the routing table entries for the originator in all nodes traversed by the RREQ (reverse path)
- The destination SQN is used to update the routing table entries for the destination T in all nodes traversed by the RREP (forward path)
- RREQs contain the last known SQN of the destination
- Intermediate nodes reply with a RREP (to a RREQ) if they already have a routing entry for the destination which has a higher destination SQN (e.g. generated by the RREQ from a third node)
- Whenever a RREQ received by a (intermediate) node has a higher source SQN than the one in the routing table, the entry in the routing table is updated
- Whenever a RREP received by a node has a higher destination SQN (the address of the source of the RREP!) than the one in the routing table, the entry in the routing table is updated

AODV: Route Maintenance – Principles

- Reverse/forward path entries are discarded after timeout
 - Soft state approach
- Entries are “refreshed” if data is transmitted on a route
 - Reset of timers
 - Optional hello messages emitted by nodes to check if next hop is still available
- Detection of link failures (on single hop):
 - Absence of link layer ACKs for forwarded data
 - Absence of hello messages
- Global repair of link failures: sending a **Route ERRor (RERR)** message
 - Forwarded to sender of data (route originator or destination)
 - Sender initiates new route discovery with RREQ
 - RREQ contains new SQN of the originator
 - RERRs could be unicast or broadcast (depending on the number of precursor entries associated with the respective routing table entries for invalid routes)

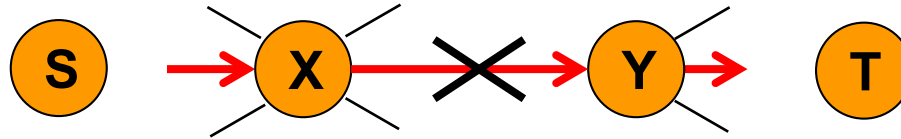
AODV: Route Maintenance



2 options to repair a broken link:

- **Global repair**
 - issuing RERR to the source S containing all destinations invalidated by broken link
 - applied when broken link is close to source or after local repair fails
- **Local repair**
 - issuing RREQ to destination T from intermediate node X where forwarding is broken
 - applied when the broken link is close to the destination

AODV: Route Maintenance – Local Repair



RREQ initiation for local repair

- Node X learns about link failure from missing ACKs or Hello messages
- Node X initiates **RREQ** broadcast with T as destination and X as originator, with X's SQN
- Node X invalidates all destinations in its routing table with Y as next hop (marks hop count as infinity)

Intermediate nodes

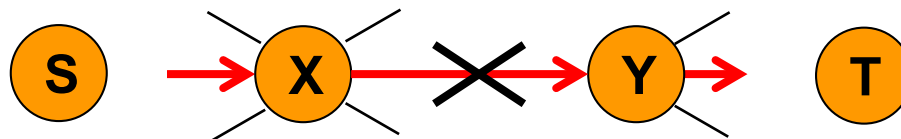
- RREQs are handled and broadcast by intermediate nodes as before

Destination node T

- Answers RREQ with RREP with X as originator

If node X does not receive a RREP from T within a given time, it initiates a RERR to S (global repair procedure)

AODV: Route Maintenance – Global Repair



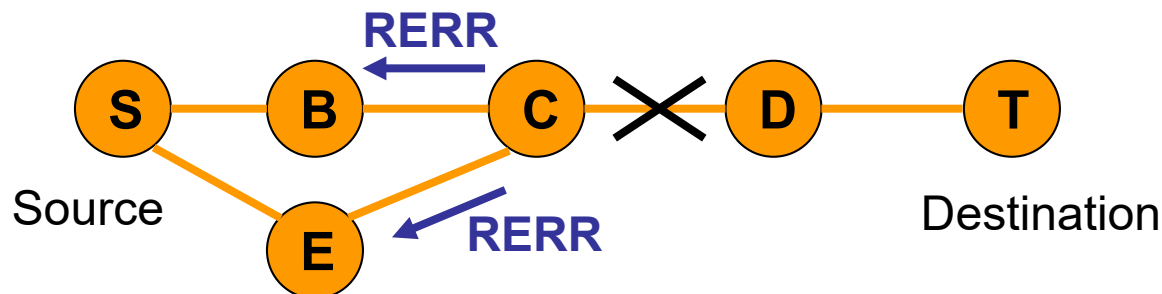
Route ERRor (RERR) message initiation

- Node X learns about link failure from missing ACKs or Hello messages
- Node X initiates RERR (broadcast or unicast) towards precursor nodes
- RERR contains all destinations in the routing table that are now unreachable (all destinations with Y as next hop), including their SQNs
- Node X invalidates all destinations in its routing table with Y as next hop (marks hop count as infinity)

Receipt of RERR (by any node)

- Invalidate all destinations in routing table (hop count – infinity) that are part of the RERR (unreachable destinations)
- If a routing entry has been invalidated, forward RERR (broadcast/unicast)

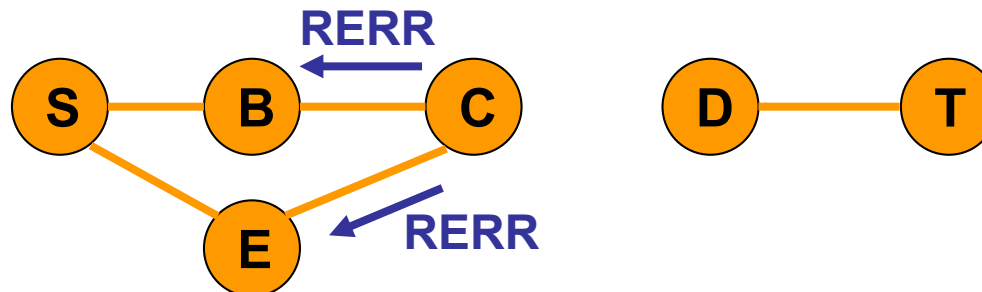
AODV: Route Maintenance – Global Repair Example



Initiation of RERR for global repair by C

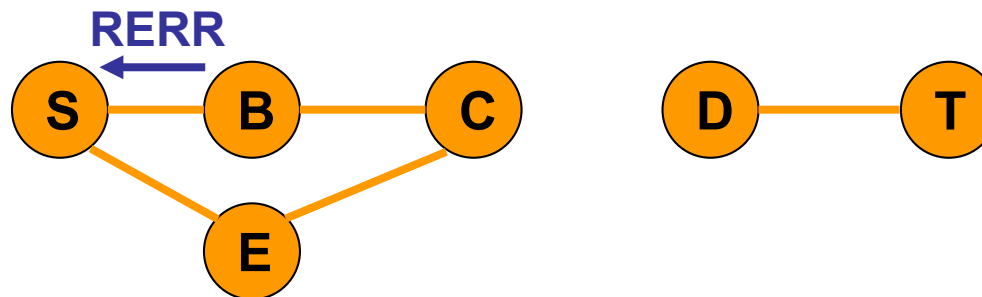
- Link between C and D breaks (lack of ACK or Hello messages)
- Node C invalidates the route to T in its routing table (and all other destinations with D as next hop)
- Node C creates RERR
 - Contains T and all destinations that are now unreachable (i.e. all entries in its routing table employing D as the next hop), including their SQN
 - RERR is broadcast to all nodes in transmission range (unicast if there is only one precursor node in the routing table)

AODV: Route Maintenance – Global Repair Example



- Node B receives RERR
 - B checks whether C is its next hop on route to T (and all other unreachable destinations given in RERR)
 - As C is next hop for B on route to T (and possibly others)
 - B deletes route to T (sets hop count to infinity)
 - B broadcasts RERR
- Node E receives RERR
 - E checks whether C is its next hop on route to T
 - As C is not next hop for E on route to T (or any other destination given in RERR), no further action by E (broadcast terminates)

AODV: Route Maintenance – Global Repair Example



- Node S (sender of initial data) receives RERR from B
 - Checks whether B is its next hop on route to T (or of any other destination given in RERR)
 - Node S deletes route to T from routing table
 - Node S rediscovers new route to T with RREQ if still needed (i.e. if it still has data to transmit to T)

AODV: Loop prevention

2 issues:

- (1) Loops in route discovery process
- (2) Loops in data traffic due to wrong routing table entries

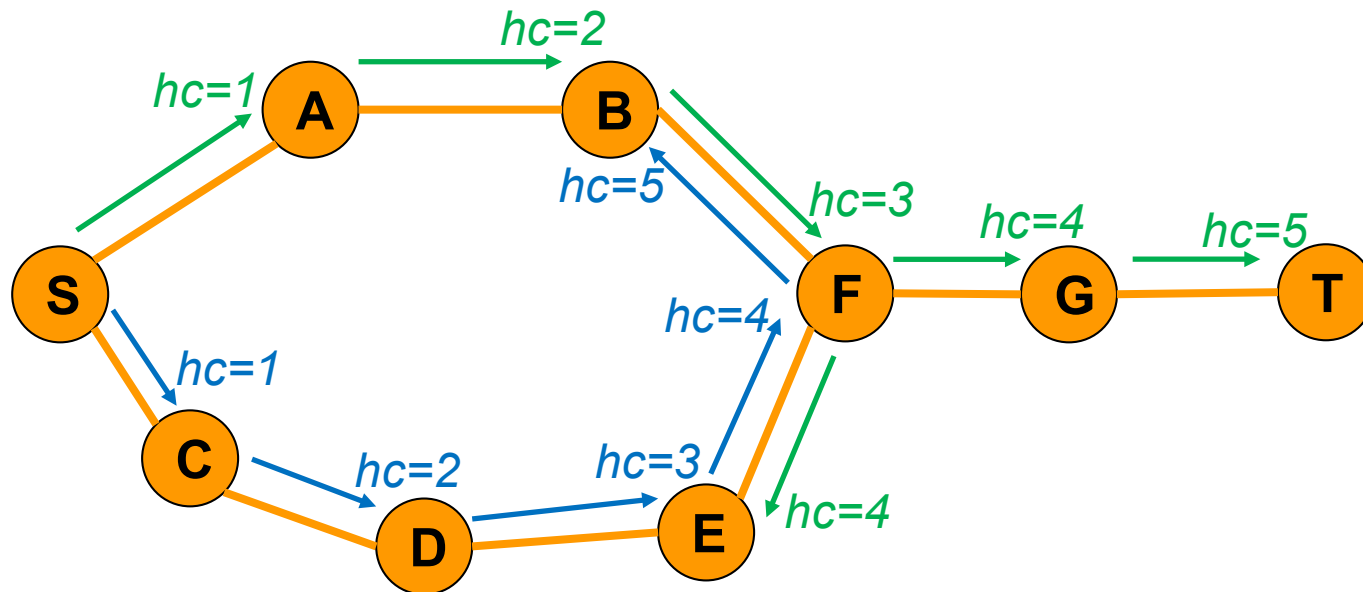
Case 1: Loops in the route discovery process

- Hop count ensure that duplicates are eliminated (limits flooding)
- Time-to-live (TTL) limits the diameter of the route discovery
- SQN prevents confusion between old and new RREQs (freshness)
- Route discovery terminates after all nodes of the network within the diameter of TTL have a route established to the originator
- New SQN indicates a change of the topology requiring an update of the routing tables
- Unicast RREP generates a single bidirectional route between S and T

Case 2: Loops in data traffic due to wrong routing table entries

- Unicast RREP using a unique SQN ensures a unique route for any established route from node A to node B in the network

AODV: Loop prevention during route discovery



- Network nodes may receive multiple RREQs, e.g. via loops, for the same source node SQN
- A node forwards a RREQ and updates its routing table only if:
 - The RREQ has the same source SQN and lower hop count (*hc*)
 - Nodes B and E discard subsequent RREQs due to higher *hc*
 - Node F forwards RREQ with lower *hc*
 - The RREQ has a higher (newer) originator SQN (update irrespective of *hc*)

AODV: Summary of issues to be solved

- Reception of multiple RREQs by a node
 - Forward/broadcast the first received RREQ only, unless a subsequent RREQ indicates a shorter route (indicated by HopCount) to the source node given in the RREQ
- Handle outdated routing information, e.g. due to mobility of nodes
 - Routing information in nodes is implemented as “soft state”
 - SQN indicates a new round of route discovery
 - Route lifetime in routing table indicates freshness of route
 - “Auto refreshing” of lifetime using data or control messages
- Limit flooding of the network with RREQs
 - TTL in RREQ limits the diameter of the network discovered
 - No forward of replicated RREQs that do not indicate a shorter route
 - RREP is unicast!
- Loops caused by inconsistent distributed routing information
 - Prevented by maintaining HopCount and SQN for discovery process in RREQs and routing tables

Comparison between OLSR (proactive) and AODV (reactive)

Protocol Property	OLSR	AODV
Approach	Proactive	Reactive
Link support	Unidirectional	Bidirectional
Control overhead	Periodic broadcasts in entire network, independent of number of used routes	Broadcasts only for active routes
Data delivery latency	Low for small amount of data, as routes are predetermined	High for small amount of data due to on demand approach
Scalability	Poor in large networks due to increased link status updates	Good, requires only active link status updates
Mobility	Requires higher periodicity of routing broadcasts	Route maintenance due to mobility only for active routes

Best approach depends on specific requirements and needs of the ad hoc network

⇒ lots of optimizations are possible for specific applications and network details

References

Recommended Reading

- J. Schiller: “Mobile Communications” – Chapter 9, Addison Wesley, 2003.
- S. Basagni, M. Conti, S. Giordano, I. Stomjmenovec: “Mobile Ad hoc Networking”, A John Wiley & Sons, Publication, 2004
- Perkins, et. al: Ad hoc On-Demand Distance Vector (AODV) Routing, RFC 3561, July 2003
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Further Reading

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